



SMART ZIMBABWE 2030 MASTER PLAN

ICT TRANSFORMATION FOR A SMART ZIMBABWE

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1. Executive Summary

Zimbabwe, like many other countries globally, is fully aware of the importance of information and communication technologies for socio-economic development and transformation. It is acknowledged that those countries from both developed and developing world who recognized the potential of information communication technologies (ICTs) and who went on to develop policies and implementation strategies, and invested in ICTs have realized immense benefits for their economies and their peoples. Zimbabwe has great potential and leveraging on its high literacy level, strong infrastructure base (communication, energy and road networks), geographical location and favorable climatic conditions, can leapfrog and overtake the developments made by some countries, particularly in Africa.

The Government of the Republic of Zimbabwe is highly committed to streamlining ICTs because they are cross-cutting and enablers across all sectors of the society and economy. Admittedly, there are some challenges we have to overcome and these include inadequate infrastructure (communications and electricity), cost of access to ICT devices, internet costs and interoperability challenges among others. As we enter a new era of advanced ICTs technological developments characterized by fourth Industrial Revolution and fifth generation technologies where all things are sensing, connected and intelligent, countries which do not embrace these technologies will be left out and will lag behind and most likely be dominated over by those who will have invested in new technologies. To systematically exploit the potential of ICTs for national development and transformation, Zimbabwe needs to develop an all-inclusive guideline that clearly articulates how the country will develop, deploy and manage ICTs across all sectors. We need to develop a Smart Zimbabwe 2030 Master Plan.

The Smart Zimbabwe 2030 Master Plan will work through key specific sector-focused integrated *Pillars* that act as Smart Solutions and these include: a) Smart Government, b) Smart Cities, c) Smart Agriculture, d) Smart Education, e) Smart Transport, f) Smart Health, g) Smart Tourism, and h) Smart Mining, among others. These Pillars are anchored on the following strong foundation or common *Platforms*:

- Policy, Regulation and Standards;
- Secure and Shared Infrastructure;
- Partnerships, Skills, Capacity Building and Content Development;

- Confidence and Security of networks and services.

The operationalization of the Smart Zimbabwe 2030 Master Plan calls for a new paradigm shift where Government Ministries will be critical in the development of programmes and projects for specific Smart Solutions for sectors they superintend over. This document presents models of Smart Government, Smart Education and Smart Cities as examples of the key tenets in a Smart Solution environment. *Coordination of the adoption, development and deployment of ICT solutions and applications should be properly managed under the guidance of the Ministry of ICT to avoid duplication and ensure optimal utilization of financial and human resources.* As the country moves towards becoming an Upper middle-income society by 2030, we need to adopt a completely new approach and exploit the potential of ICTs by developing appropriate and cost-effective applications that can improve the country's productivity and competitiveness. Various ICT applications and solutions can improve industrial processes and quality of products, increase agricultural yields, enhance teaching and learning, positively impact on health, governance among many other areas in the society. With systematic adoption and implementation of Smart Zimbabwe 2030 Master Plan, the country will be transformed into a knowledge-based and fully inclusive Smart Society by year 2030. For this reason, the Ministry of Information Communications Technologies, Postal and Courier Services presents here a comprehensive, focused, forward thinking approach on how Zimbabwe can exploit the potential of ICTs for socio-economic development and transformation. The Smart Zimbabwe 2030 Master Plan proposes that the Ministry of ICT, Postal and Courier Services becomes the focal point for all ICT projects and programmes. The Ministry should coordinate ICT programmes in Ministries and ICT Departments/Units should be linked to the Ministry similar to Finance and Legal Departments in Ministries which are linked to Accountant General and Attorney General's Office respectively.

This **Sectoral Master Plan** is also critical for the development of inter-linkages with the **National Integrated Infrastructure Master Plan**, which are currently being developed under the Ministry of Higher and Tertiary Education, Transport and Infrastructural Development and the Office of the President and Cabinet.

2. Background

The uptake and use of ICTs has greatly increased in recent years resulting in considerable reduction in the “digital divide” between inter-urban, rural and urban areas. The Inter-Census Demographic Survey (ICDS) conducted by the Zimbabwe National Statistics Agency (ZIMSTAT) from 18 to 23 August 2017, which was officially launched in May 2018, shows that the country’s literacy rate stands at 94 percent - the highest in Africa and this, in a way, has contributed to the increased uptake of ICTs in the country. This growth is evidenced by Zimbabwe’s ICT indicators where mobile penetration has reached 93.1% and internet penetration was at 62.9% as at 31 December 2018. The national 2G and 3G coverage stands at 93% and 84% respectively against the regional survey of 92% and 73%. Zimbabwe stands at 35% against a regional average of 42% in respect of LTE Population coverage.

Rapid and robust infrastructural development and rejuvenation has enabled the development and availability of a plethora of e-services, which consumers have embraced as easier means to communicate and transaction between person to person, person to business and business to business, government to government and various e-Government services to the citizenry. There has been an increase in the adoption of mobile money transfer, mobile wireless broadband, use of plastic money and various social applications such as WhatsApp, Facebook, Twitter, You Tube and Skype, to name but a few.

Zimbabwe is now connected to the undersea optic fibre cable networks through SEACOM and EASSy cables in the Indian Ocean and WACS cable in the Atlantic Ocean. The efforts from the public and private sector operators in this development are commendable.

The SMART Zimbabwe 2030 Master Plan seeks to transform the country through use of ICTs so that Zimbabwe attains its Vision of becoming an Upper Middle-income economy by 2030. This Vision is predicated on, among others, building on the achievements of the Zimbabwe National Policy on ICTs of 2016 to 2020 which further strengthens Zimbabwe’s economic base and improve its economic environment for accelerated growth towards achieving a predominantly information and knowledge-based economy by 2030. The current National ICT Policy now requires a review and re-orientation for it to be aligned to the new initiatives of SMART ICT as defined by the SMART Africa Manifesto of the Transform Africa Summit. Zimbabwe has just become the 26th Member of Smart Africa Alliance where it adopted Smart Agriculture as its flagship focus area thus opening doors for its contribution to the Africa Preferential Trade initiative. Such efforts will result in the formulation of a new National ICT

Strategy, the SMART Zimbabwe 2030 Master Plan, which will guide Zimbabwe as it streamlines and accelerates the adoption of digital technologies.

3. ICT Sector Challenges in Zimbabwe

Despite the developments and potential that are outlined in 2 above, the ICT sector has been daunted with a number of challenges which include the following: -

- i) Inadequate communications infrastructure*
- ii) Inadequate Electricity and road networks*
- iii) Affordability of smart devices and other ICT products*
- iv) Inadequate ICT skills*
- v) Low digital literacy level*
- vi) Inadequate investment capital*
- vii) Limited local ICT innovation, R&D and entrepreneurship*
- viii) Interoperability challenges*

There are no cross-network transactions for digital financial services and the country has not yet reaped the advantages of cross-number portability. There is also no interoperability between data centres, systems and databases that currently exist across Ministries and public sector institutions.

4. The SMART Zimbabwe 2030 Master Plan

Having realized the importance of ICTs and their potential to transform the country into a knowledge-based, smart and fully inclusive society, the Government of Zimbabwe has developed the SMART Zimbabwe 2030 Master Plan and Implementation Strategy, which is inclusive and cross-sectoral. The Master Plan identifies and prioritizes clear programs, projects and targets which the country should vigorously pursue while driving towards realizing the National 2030 Vision. Treasury and the Public Service Commission are expected to provide adequate financial and skilled human resources to ensure successful implementation of SMART Zimbabwe 2030 programs and projects. Public, Private sector

Partnerships should be encouraged in order to realise the aspirations of Smart Zimbabwe 2030 goals. The Master Plan also puts in place a strong and well-coordinated institutional arrangement which is inclusive and has the mandate to efficiently coordinate ICT interventions in all sectors of the economy through the Ministry of ICT, Postal and Courier Services. The Smart Zimbabwe 2030 Master Plan contains inputs from the wide spectrum of extensive stakeholder consultative meetings held over time. The Smart Zimbabwe 2030 Master Plan thus serves as an inclusive roadmap for the country's comprehensive and stakeholder supported development and transformation initiative through ICTs.

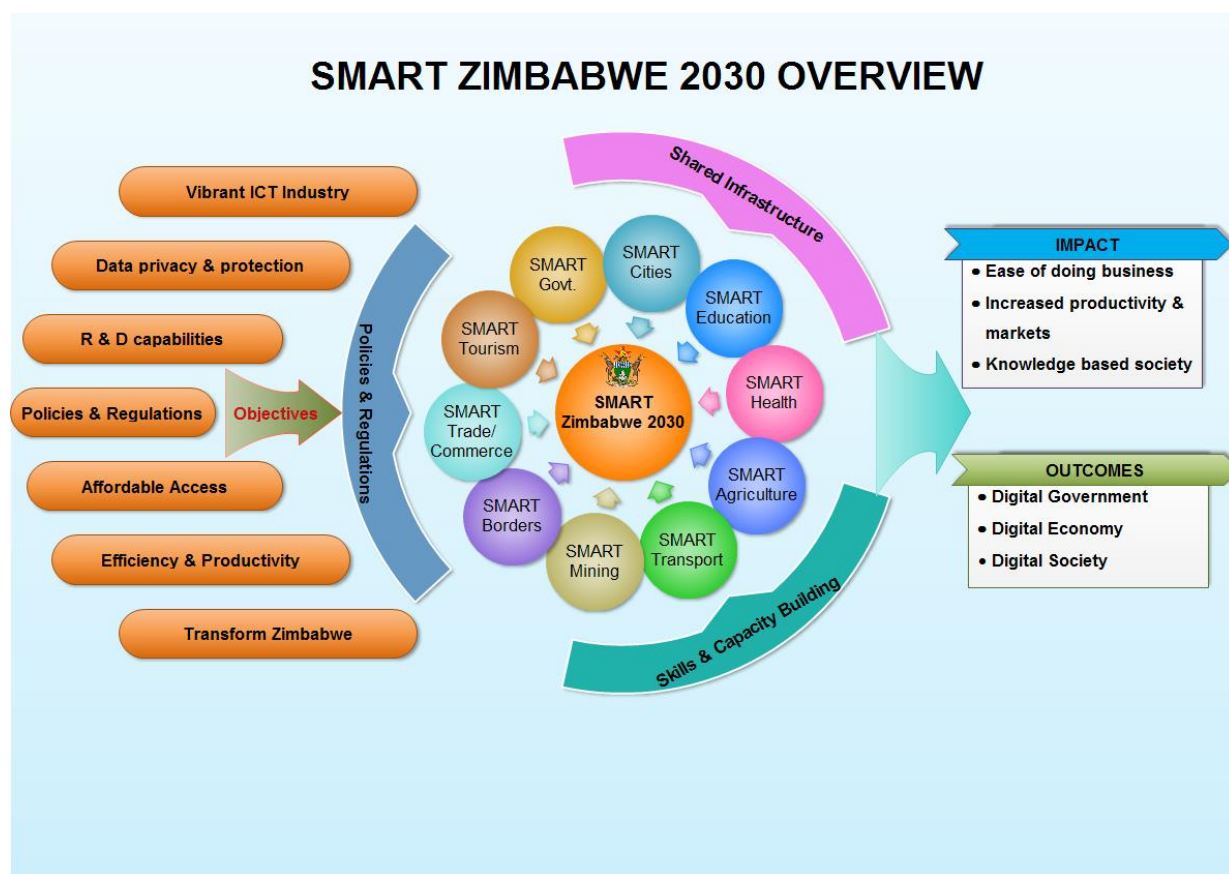


Fig. 1 SMART Zimbabwe 2030 Master Plan Overview

4.1. Vision and Mission

4.1.1. Vision

"To create a knowledge-based and digitally inclusive society and a globally integrated economy by 2030."

4.1.2. Mission

“To fully integrate information and communications technologies throughout society and into all sectors of the economy in order to realize rapid and sustainable socio-economic development.”

4.2. Goals

The overall SMART Zimbabwe 2030 goals are to: -

- i) Increase wealth creation and improve quality of lives through the adoption and use of ICTs;
- ii) Promote affordable, universal access to ICT products and services;
- iii) Promote effective regulation of the sector by establishing rules, practices, regulations and standards that promote competition, protect and educate the consumers, create a level playing field among operators and service providers and encourage local and foreign investment in the ICT sector;
- iv) Increase ICT and digital literacy levels in the country;
- v) Provide mechanisms for empowering local participation in the ICT sector;
- vi) Develop a vibrant ICT industry that creates employment and contributes to national economic development and position the ICT industry as one of the key exporters;
- vii) Transform Zimbabwe into a Regional Hub through the development of innovation hubs that produce goods and services that are competitive nationally and in the region;
- viii) Enable Government, organizations, business and individuals to become more efficient and productive through ICTs; and
- ix) Establish and maintain information and communications technology data privacy and personal data protection.

4.3. SMART Zimbabwe 2030 Structure

SMART Zimbabwe 2030 is therefore a bold and innovative commitment/initiative of the Government of Zimbabwe to accelerate sustainable socio-economic development of the country thereby ushering Zimbabwe into a knowledge-based economy by 2030 through affordable access to broadband and usage of ICTs. The initiative relies on several Pillars

or ‘enablers’ that guide achievements of a range of economic and social objectives all aimed at Zimbabwe attaining an increasingly digitized and knowledge-based economy.

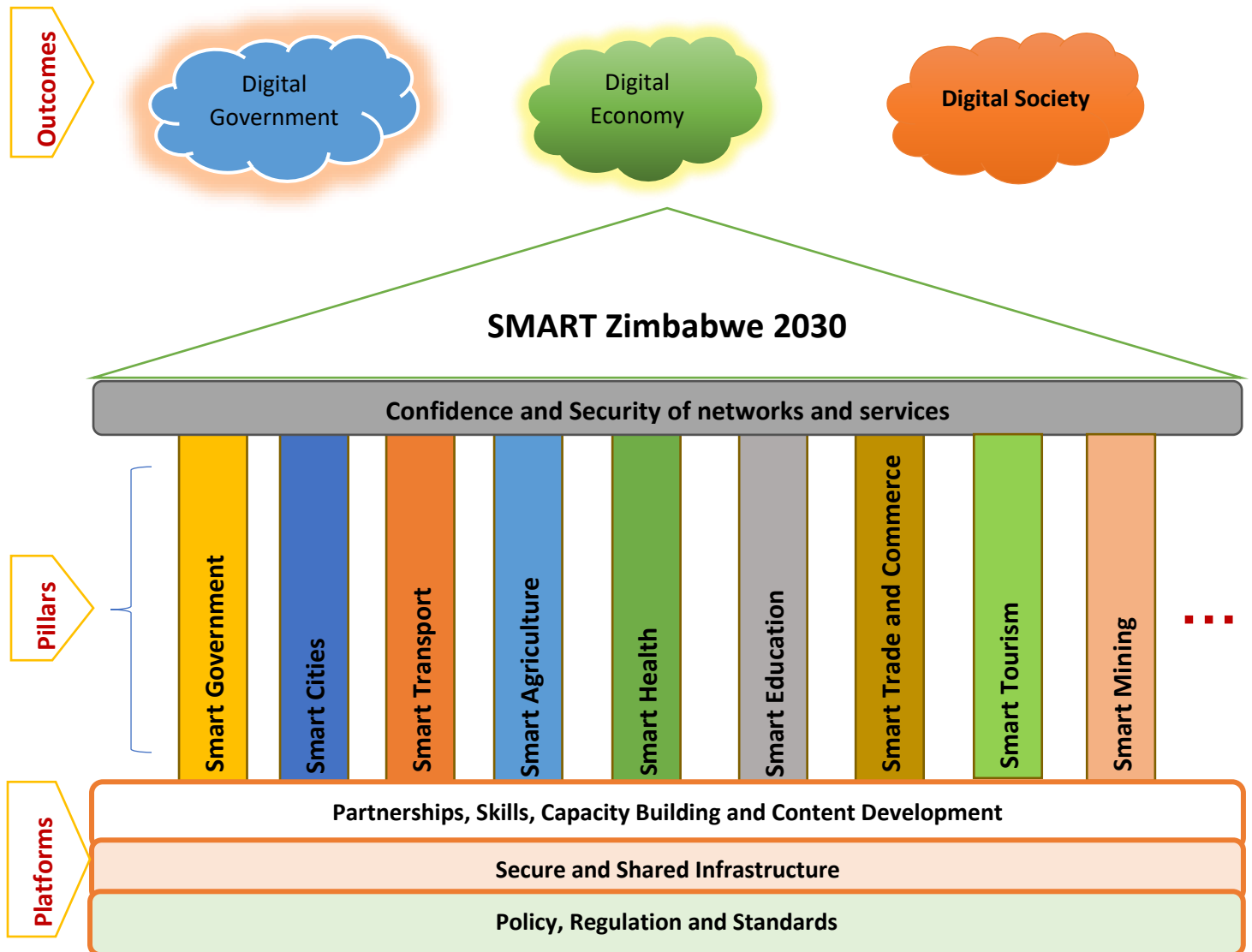


Fig. 2. SMART Zimbabwe 230 Structure

4.3.1. SMART Zimbabwe Common PLATFORMS

SMART Zimbabwe Master Plan can be likened to a structure that is solidly anchored on three common Platforms which act as the foundation on which the various SMART Pillars/Solutions are rooted on. The common Platforms described below should constantly be reviewed to ensure their alignment to technology and skills trends.

4.3.1.1. Policy, Regulation and Standards

The enabling policy and regulatory environment platform will initiate long-term support for integrated national ICT policy and standards reviews for sectors and the country, to assess the overall status of ICT sector reform and market liberalization, and support national broadband strategy development. The Policy, Regulatory and standards should also respond to regional dictates.

4.3.1.2. Secure and shared Infrastructure

All ICTs work through and on a secure, reliable and expansive communications infrastructure nationwide. The infrastructure comes in various forms and configurations that include hardware, Local Area Networks, Wide Area Networks, last mile connectivity all linked to Optic fibre, Radio, VSAT, Mobile and Fixed Communications network solutions. As a land-locked country, Zimbabwe is connected to the undersea optic fibre cables through Mozambique, South Africa and Namibia.

ICTs need good road networks countrywide and for ICTs to work, there is need for reliable and secure electricity supply and therefore provision of electric power should be available countrywide.

4.3.1.3. Partnerships, Skills, Capacity Building and Content Development

Zimbabwe has a high literacy level and should leverage on this to become a software and hardware development hub in Africa. The Government of Zimbabwe intends to increase innovation through promoting local content, heritage and cultural applications development. This Platform is aimed at complimenting private sector efforts to bridge the digital divide and empower women and young Zimbabwean entrepreneurs to participate in the creation of locally relevant content and applications that will increase Zimbabwe's web presence and build a software industry. One of the strengths of ICTs is the way they can assist unlock distant expertise, knowledge and markets through use of Zimbabweans in the diaspora. Local content should also be perceived as a driver for local job and wealth creation. It is also a potential foreign currency earning industry through the export of locally developed software and applications. There should be a deliberate move to develop content that is consistent with linguistic and cultural diversities in the country.

4.3.1.4. *Confidence and Security of Networks and Services*

There will be need for reliable and secure communication networks by establishing Computer Emergency Response Teams and standards for fibre optic cable laying and topologies in order to minimize downtimes. Given the heavy investment in the ICT infrastructure to support the economic development goals, it is imperative that infrastructure be resilient and secure against cyber threats as this will protect systems, citizens and applications that use the infrastructure.

4.3.2. The PILLARS of SMART Zimbabwe 2030 Master Plan

In order to achieve SMART Zimbabwe 2030, several SMART Solutions focusing on specific sectors should be developed and implemented. These Smart solutions should be developed working with sector Ministries and stakeholders to ensure that the solutions are comprehensive and there is enough buy-in and collaboration to ensure effective implementation. The following examples demonstrate the frameworks that can be further developed to form Smart Solutions that address Smart Government, Smart Cities and Smart Education.

4.3.2.1. Smart Government

Smart Government is the use of innovative policies, business models, and technology to address the financial, environmental, and service challenges facing public sector organizations. The concept of Smart Government relies on consolidated information systems and communication networks that expands countrywide. Smart Government is about using technology to facilitate and support better planning and decision making. Smart Government is also about improving democratic processes and transforming the ways that public services are delivered. Smart Government therefore is a new way of governance relying on information and communication technologies and it is citizen centric, data driven and performance focused.

Find Fig 3. below shows the diagrammatic structure of SMART Government;

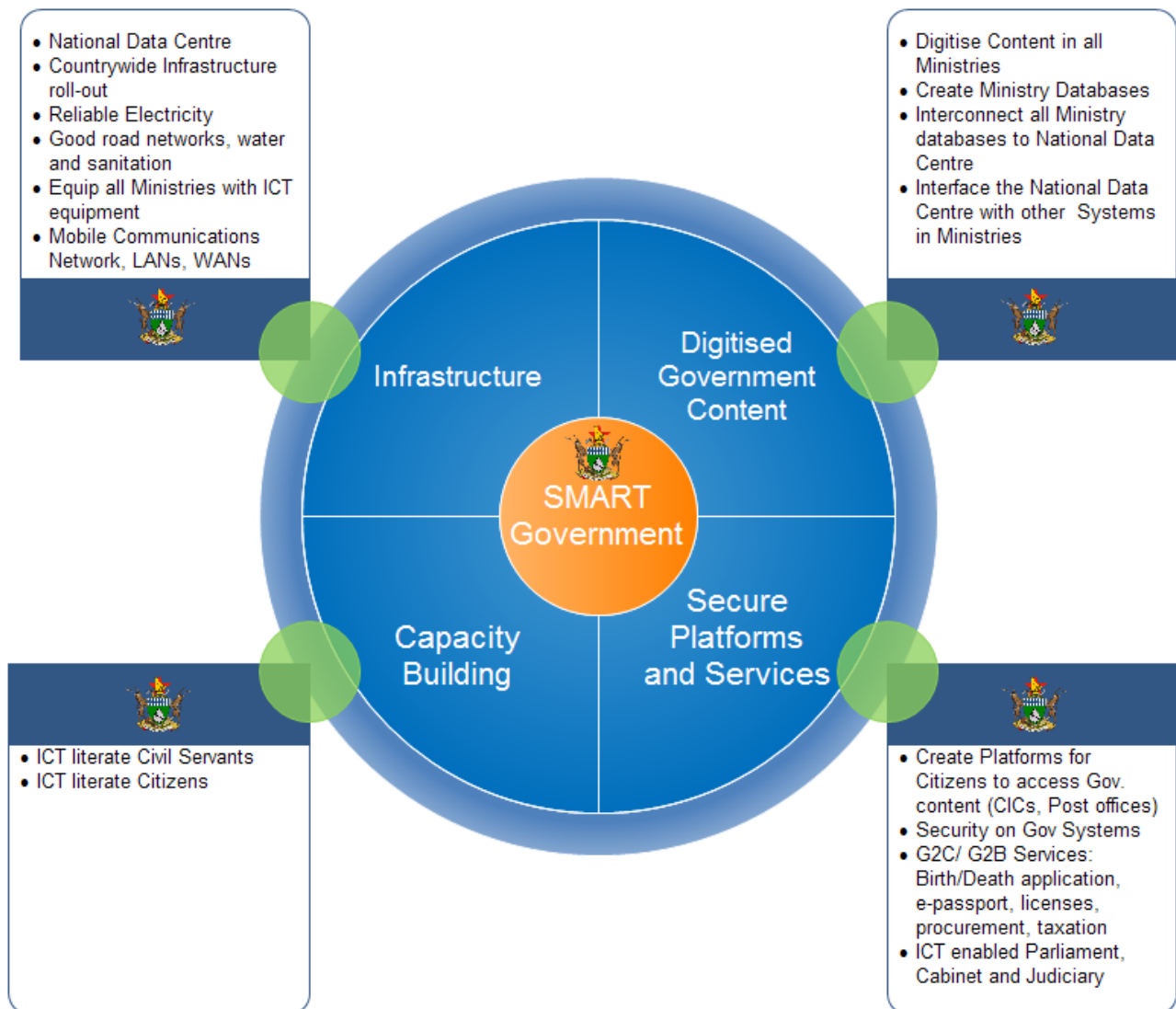


Fig. 3 SMART Government Overview

The Smart Government Pillar will result in reforming the Government of Zimbabwe through ICTs. The Pillar covers all forms of digital interaction between Government and its citizens. The strategies employed can be divided into the following Six (6) key areas:

a) Infrastructure:

Government services cover the entire country and therefore there must be comprehensive, reliable and secure communications infrastructure. We also need road networks and electricity supply that reach all corners of the country.

b) Digitised Government content

All Government data and information which currently is predominantly file and paper-based should be digitised and systematically stored in electronic format.

c) Platforms and Services

Create platforms where Civil Servants and other citizens will have access to Government content and services. These platforms should also provide Citizens with opportunities to participate on governance issues.

d) National Data Centre

All data and information from Government and sectors of the economy should be stored in databases that are networked and linked to the National Data Centre facilities.

e) Capacity Development

SMART Zimbabwe requires skilled manpower who are equipped with skill-sets to manage and operate modern equipment and systems. Zimbabwean Citizens must attain high ICT literacy so as to optimally utilise 21st Century facilities and services.

f) ICT Security

Security should be embedded at all levels of operations and services to protect Citizens and systems against cyber threats.

Key stakeholders

- Office of The President and Cabinet
- Ministry of ICT, Postal and Courier Services
- Government Ministries, Parastatals and State Owned enterprises
- ICT Sector Players
- Business Communities
- Citizens

Benefits

- Improved service delivery
- Efficient Government that is responsive to the needs of Citizens
- Paperless Cabinet
- Efficient management of Government resources (financial, personnel and assets).
- Participation in governance issues by Citizens
- Improved communication within Government and between Government and Citizens.

4.3.2.2. Smart Education

The Government of Zimbabwe intends to increase ICT usage in primary and secondary schools, and tertiary institutions to enhance teaching and learning. The success of SMART Education hinges on the provision of connectivity and infrastructure (electricity, road networks and ICT equipment), Skilled and well-equipped Teachers/Instructors and appropriate content that address cultural and linguistic diversities in Zimbabwe.

We seek to achieve knowledge-based economic developments through education and human resources development and utilization. Therefore, it is necessary for Zimbabwe to adopt strategies and structures (see Fig 4. below) that ensure Zimbabwe maintains its high education level and skilled manpower development thrust.

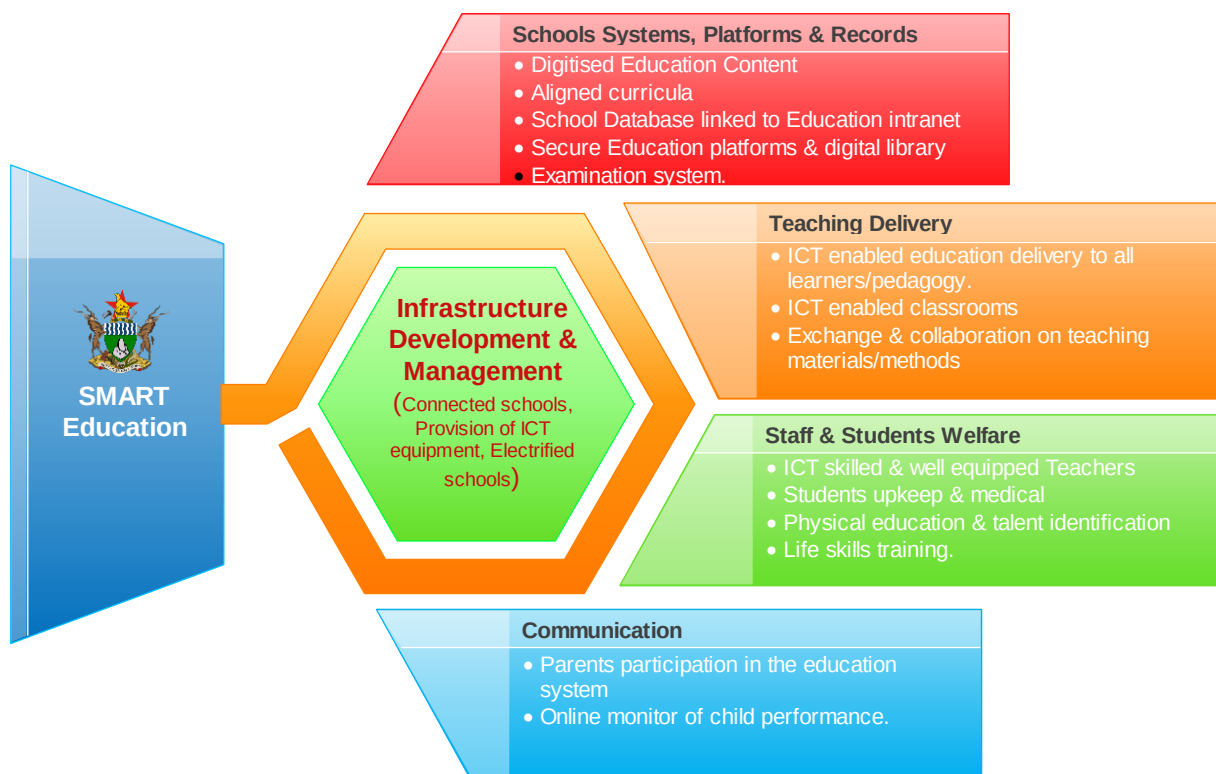


Fig. 4. Smart Education

There is need to collaborate with relevant institutions, Government Ministries, schools, colleges and universities to develop programs that increase ICT human resource capacity and skills development so as to provide equitable access to ICT enabled education and training in all parts of the country, including disadvantaged communities.

We should encourage and support research and development in ICTs and promote e-learning and use of e-learning materials throughout Zimbabwe. Smart Education should be linked and closely collaborate with technology parks in Provinces and innovation hubs at universities countrywide.

Key Stakeholders

- Ministry of Higher and Tertiary Education Science and Technology Development
- Ministry of Primary and Secondary Education
- Ministry of ICT, Postal and Courier Services
- Office of the President and Cabinet
- Other Government Ministries
- Education Sector Players

- ICT Sector Players
- Development Partners
- Citizens

Benefits:

- Transformation of the Education sector into a regional manpower development hub.
- Export of Human skills and earn revenue for the country.
- Application of education to enhance sustainable livelihoods and entrepreneurship.
- Use of schools and tertiary institutions as centres of content development solutions
- Linking of all learning institutions to the national education network with full online access to digital content in the education cloud.
- Increased education opportunities and information for ordinary citizens.
- Reduced education costs and improved service quality.
- Enhanced students/teacher interaction through open and distance learning
- Integrated students and teacher's information.

4.3.2.3. Smart Cities

A Smart City is a developed urban area that creates sustainable economic development and high quality of life by excelling in multiple key areas such as: economy, mobility, transportation, built environment, Education facilities, Health facilities, Smart Infrastructure (electricity, lighting, surveillance, road networks, water, waste management), people, accommodation (houses and offices), and government (municipalities/local authorities).

The advent of Smart Cities is one of the greatest challenges and field of opportunities in the goal to achieve sustainable, comfortable, and socially responsible living environments. A large number of factors, spanning from Government / administration / citizen interaction models, heterogeneous communication network, interoperability, or security determine the capabilities and functionalities that can be deployed resulting in a new model of public organization. Some of the key aspects that make up a Smart City solution are shown in Fig 5 below.

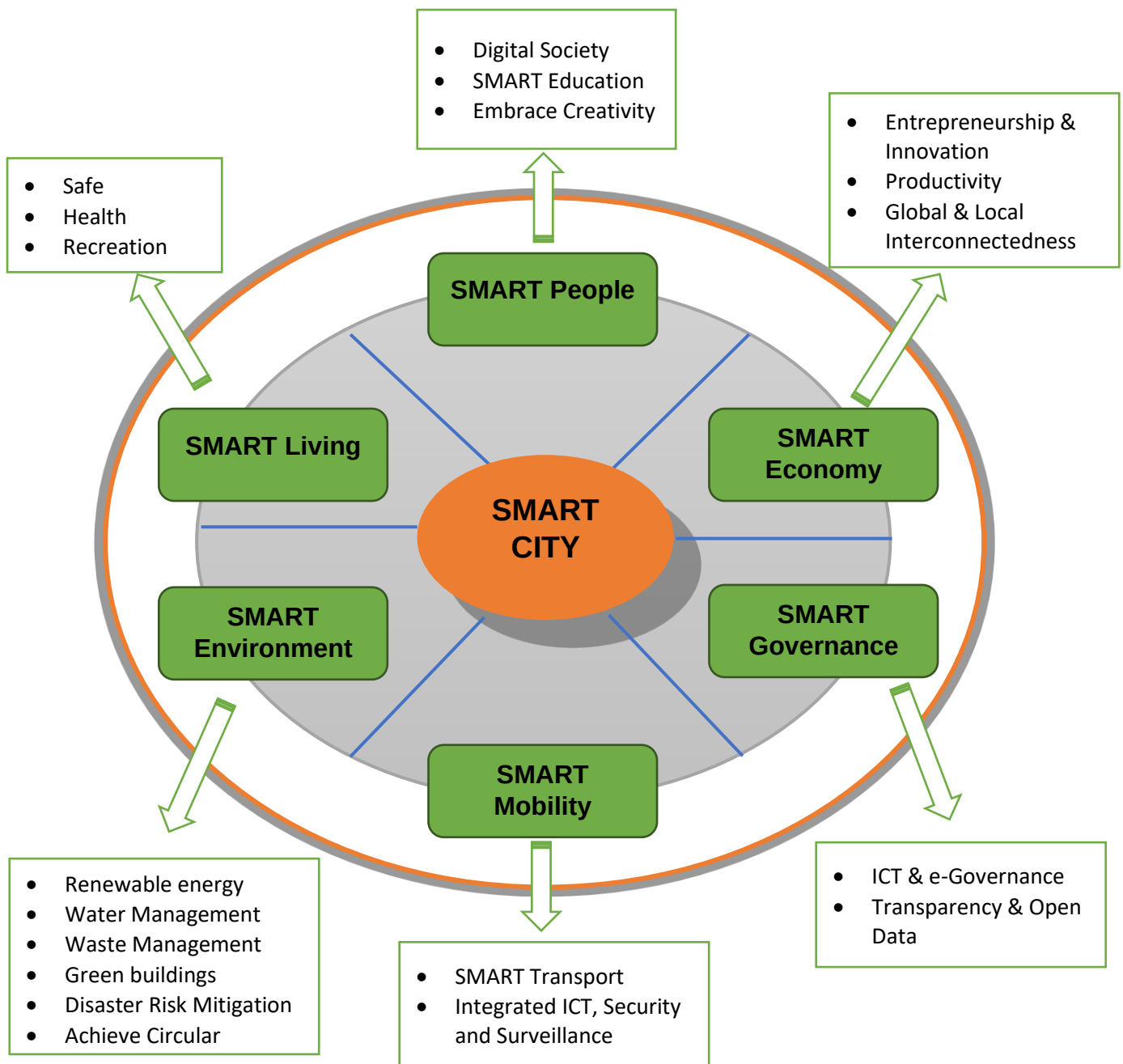


Fig. 5 SMART City Solution Overview

The SMART City solution can be further explained by Fig 6 below:

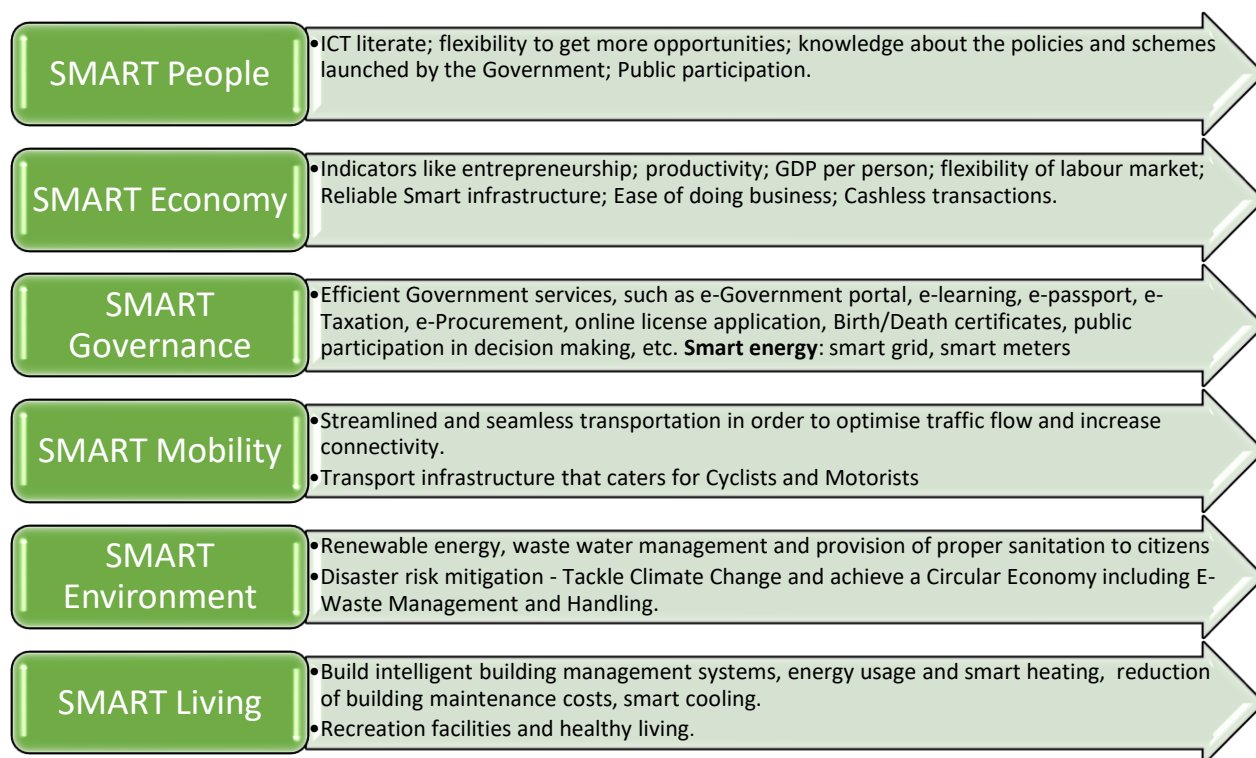


Fig. 6 SMART City Indicators

Data and Information in a smart city is collected through integrated set of sensors strategically located throughout the city and monitored through a network operating centre. The collected Citizens and City Authorities through specific and secure applications such as health, traffic management and waste management. Data and information is stored in databases and city data centers. These layers are shown in Fig 7 below.

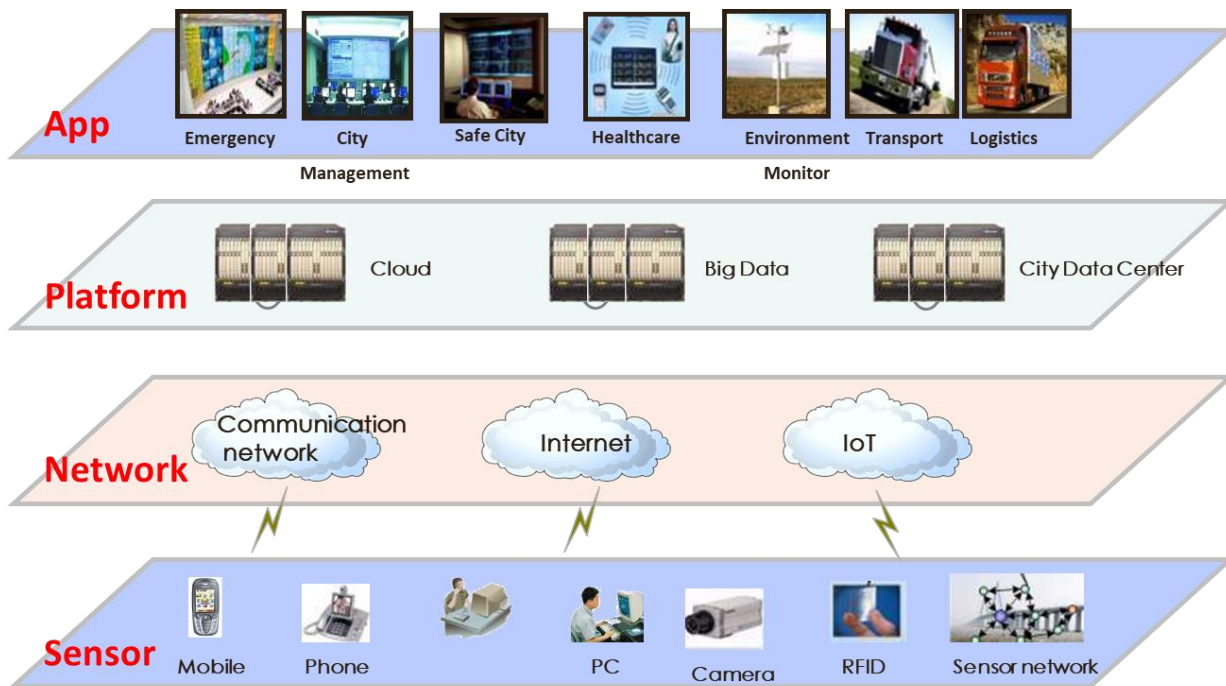


Fig.7 SMART City Layers

Key Stakeholders:

- Ministry of Local Government, Public Works and National Housing
- Ministry of ICT, Postal and Courier Services
- Office of the President and Cabinet
- Government Ministries
- Local Authorities
- ICT Sector Players
- Citizens

Benefits:

- A better quality of life for residents and visitors
- Economic competitiveness to attract industry and talent
- An environmentally conscious focus on sustainability
- Make governance Citizen-friendly and cost effective
- Better city services and a higher quality of life

4.3.2.4. Smart Agriculture

Smart Agriculture is a concept of farming management using modern Information and Communication Technologies to increase the quantity and quality of products.

Smart Agriculture seeks to enhance overall productivity in all subsectors of agriculture and provide basis for industrialization as well as the provision of timely and relevant information to the farming community.

In identifying the ways in which ICT can help agriculture, it is useful to broadly view the farming life cycle as a three-stage process:

- **Pre-cultivation:** including crop selection, land selection, calendar definition, access to credit, etc.
- **Crop cultivation and harvesting:** including land preparation and sowing, input management, water management and fertilization, pest management, etc.
- **Post-harvest:** including marketing, transportation, packaging, food processing, etc.

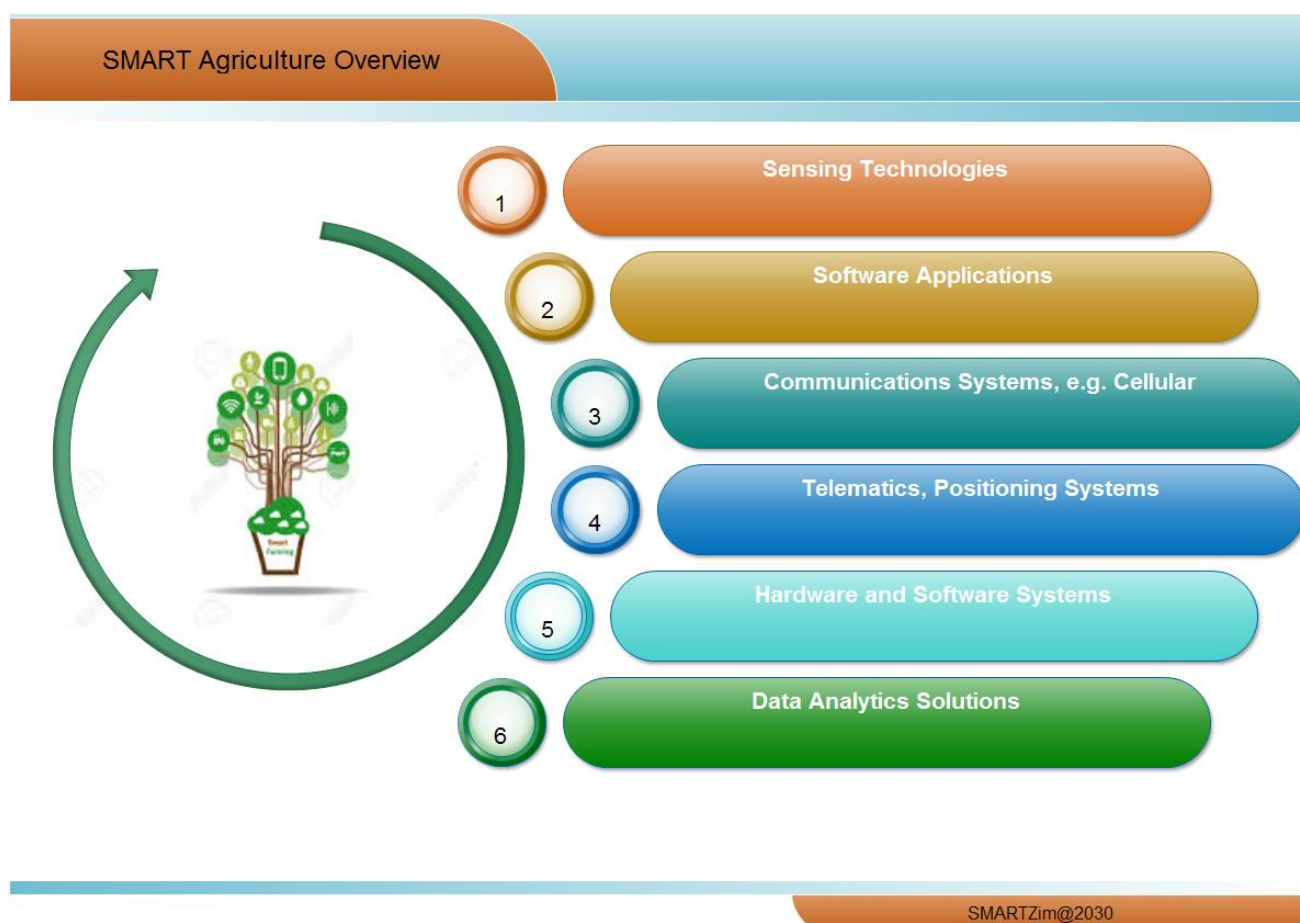


Fig. 8 Smart Agriculture

Some technologies that are available for present-day farmers include;

- ***CommSensing technologies*** - including soil scanning (type and nutrients), water and moisture, light, humidity, temperature management;
- ***Software applications*** - specialized software solutions that target specific farm types and crops suitable for agricultural regions in the country, software for livestock farming;
- ***Weather forecasting*** – precise weather prediction and dissemination of same information to stakeholders;
- ***Communication platforms*** - mobile communication access, access to markets (local, regional and international), GPS, and drone technologies;
- ***Storage technologies*** – storage technologies for all crops, grading and regrading before marketing;
- ***Hardware and software systems*** - for IoT-based solutions, robotics and automation, mechanization, drip irrigation and water conservation, and
- ***Databases and Data Centre for the agriculture sector – collection and storage of agriculture data and information, Data analytics***, agricultural inputs, dissemination of information to farmers, AGRITEX officers and decision makers.

Key Stakeholders

- Ministry of Lands, Agriculture, Water, Climate and Rural Settlement
- Ministry of ICT, Postal and Courier Services
- Office of the President and Cabinet
- Government Ministries
- Organisations and industries involved in Agriculture
- ICT Sector Players
- Citizens

Benefits:

- Improved productivity and increased efficiency in the agriculture value chain
- Diversified agriculture product portfolio

- Evolve from subsistence farming to commercial farming with export potential

5. Outcomes of SMART Zimbabwe 2030

The outcomes of SMART Zimbabwe can be summarized in three categories, namely Digital Government, Digital Economy and Digital Society. Some of the characteristics of these outcomes are captured in the diagram below (Fig 9.)

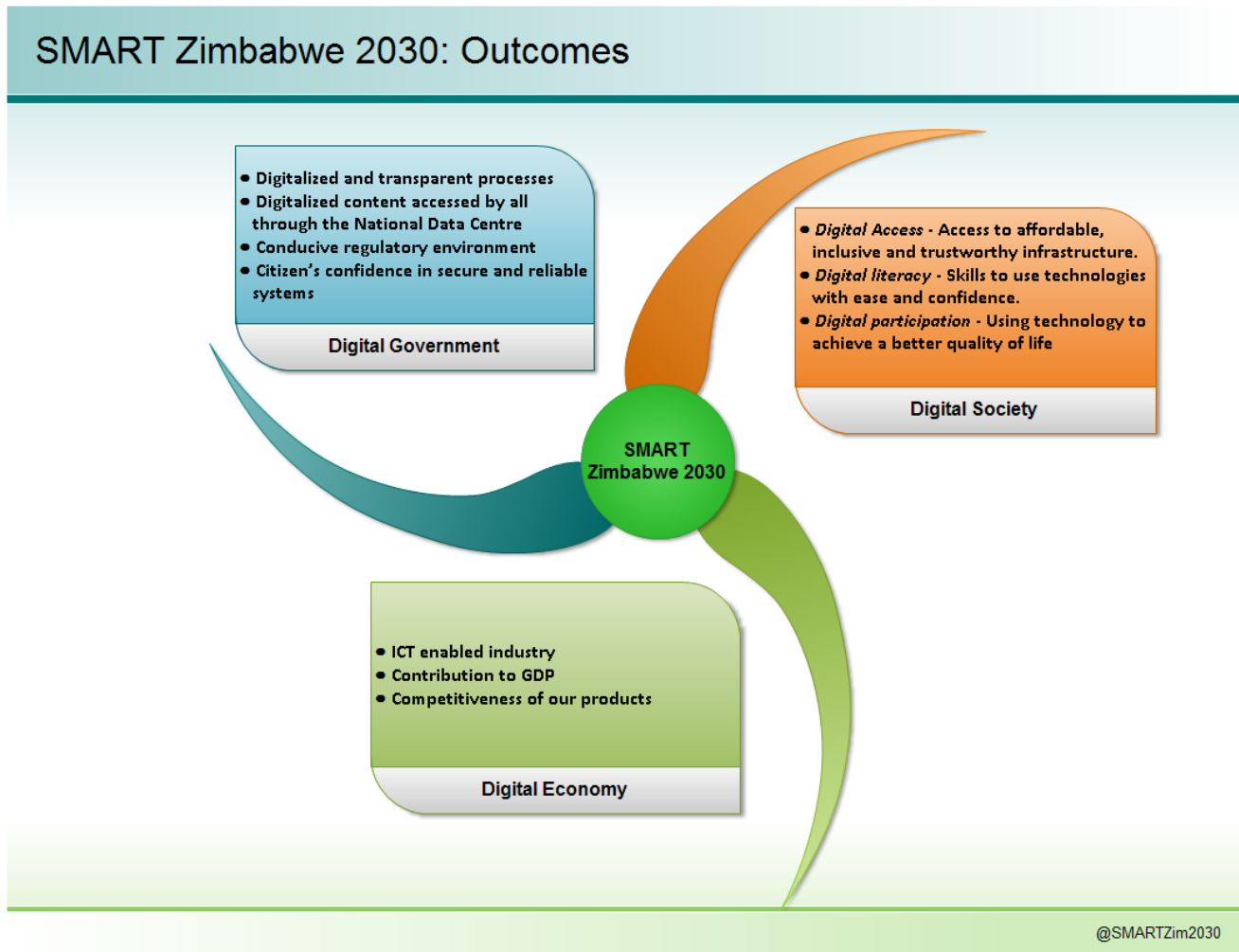


Fig. 9 SMART Zimbabwe 2030 Outcomes

6. Key Success Factors

Zimbabwe is making concerted efforts and investments in building the necessary infrastructure across the country to improve ICT access to every part of the country and achieve digital inclusion as a crucial foundation to creating a future and unified knowledge-based society. The success of the SMART Zimbabwe 2030 Master Plan will be hinged upon the following key success factors;

- i) Political will and support*
- ii) ICT Regulatory Environment*
- iii) Converged Regulator*
- iv) Availability of Financial Resources*
- v) Availability of a Highly skilled and literate workforce*
- vi) Diversified experience*
- vii) Access to the regional and world markets*
- viii) Public Private Partnership engagements*
- ix) Developed infrastructure*

7. Implementation Strategy of SMART Zimbabwe 2030 Master Plan

7.1. Smart Zimbabwe 2030 Institutional Arrangement

The proposed structure of the Smart Zimbabwe 2030 Initiative is illustrated in Figure 10 below:

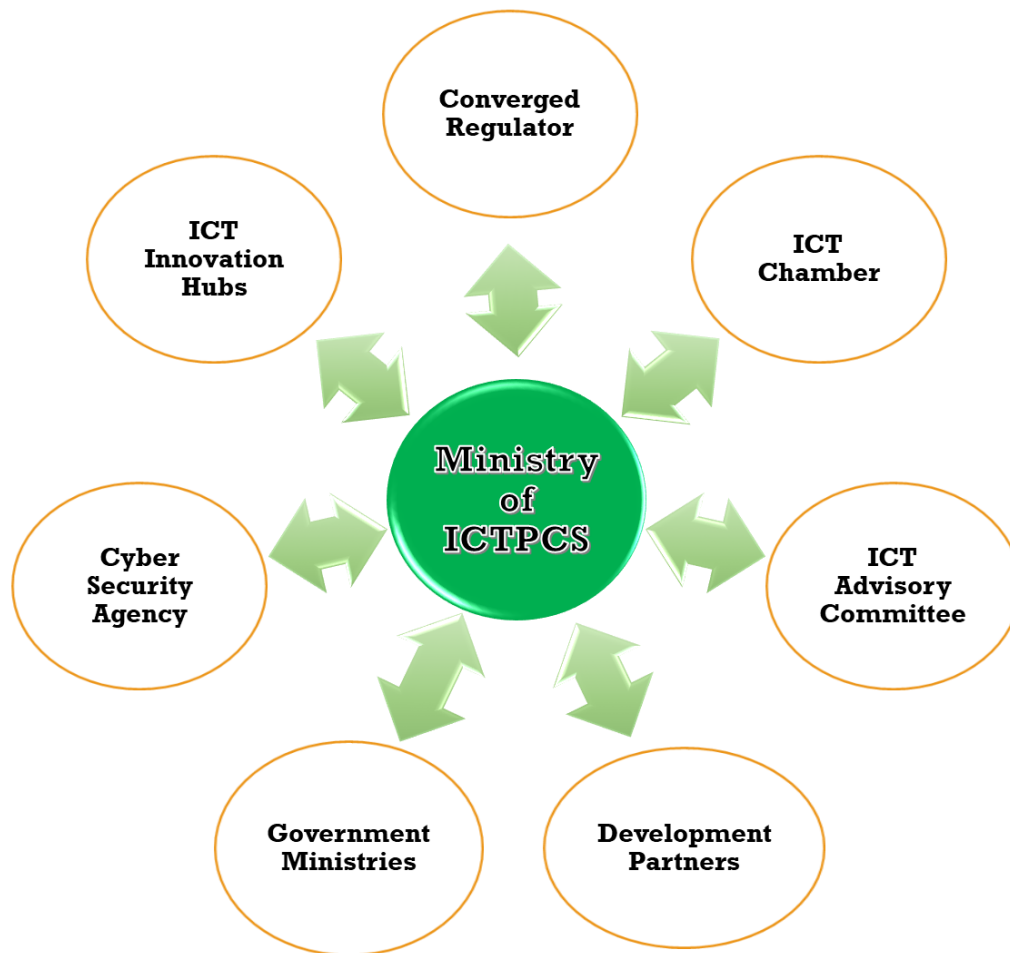


Fig. 10 Institutional Framework

As illustrated in fig 10 above, the Ministry of ICT, Postal and Courier Services acts as the central hub to which ICT sector players feed in and from which they are coordinated. This is reinforced by the fact that, elsewhere on the Continent, Rwanda's success story in the ICT Sector, for example, is attributed to the country's centralized coordination of all national ICT programmes under the country's Ministry of ICT and Innovation. As a way to enhance coordination of national ICT programmes and taking a cue from Rwanda, Government may consider critical examination of its ICT institutional framework and set up a dedicated Commission, Board or Committee under or within the Ministry of ICT, Postal and Courier Services to run with the various ICT programmes in consultation with the respective line Ministries. The following outlines the functions of the Ministry of ICT, Postal and Courier Services and the role of the relevant organs of the ICT governance structure in the country:

i) Ministry of ICT, Postal and Courier Services

The Ministry provides guidance, policy oversight and overall ICT sector coordination and leadership. The Ministry will be responsible for coordinating all ICT projects/programmes across government. All Ministries' ICT Units/Departments should be linked to the Ministry of ICT, Postal and Courier Services in the same manner the Finance and Legal Departments in Ministries are linked to the Accountant General and Attorney General's Office respectively.

ii) Converged Regulatory Authority

A converged regulatory authority (following the merger of Postal and Telecommunications Regulatory Authority of Zimbabwe and Broadcasting Authority of Zimbabwe) will promote policy harmonization.

iii) Zimbabwe ICT Innovation Hubs

Responsible for ICT skills development, content creation and innovation. These hubs will collaborate with other related hubs and technology parks in the innovation ecosystem e.g. at Government level and Ministry of Higher and Tertiary Education, Science and Technology Development.

iv) The National ICT Advisory Committee

A grouping of eminent Zimbabwean and international experts drawn together to provide strategic direction in the ICT Sector.

v) ICT Chamber

The ICT Chamber is responsible for fostering partnerships between Government and the private sector on national developmental initiatives including ICTs. The ICT Chamber will also be responsible for coordinating the activities of the private sector, civil society leading to investment and Private Public Partnerships (PPPs) in the ICT sector.

vi) National Cybersecurity Agency

An independent institution responsible for all cyber security activities and will relate to the Ministry of ICT, Postal and Courier Services and other security arms of the State.

vii) Development partners

Foreign institutions and local companies that are in partnership with Zimbabwe in its developmental initiatives particularly in the ICTs sector.

viii) Government Ministries and Departments

Implementation of ICT programs and projects in Government Ministries and Departments. These become key partners in implementing the SMART sector solutions contained in this Master Plan. This committee will ensure streamlining and full integration of ICTs across government.

7.2. Implementation plan

Delivery of the SMART Zimbabwe 2030 Strategy is broadly divided into the following three (3) phases:

	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	
Start 01/01/2020												Finish 31/12/2030
Smart Zimbabwe 2030												
Enablers												
01/01/2020 - 31/12/2030												
Policies, Regulations and Standards												
01/01/2020 - 31/12/2030												
Secured and Shared Infrastructure												
01/01/2020 - 31/12/2030												
Partnerships, Skills, Capacity Building and Content Development												
01/01/2020 - 31/12/2030												
Confidence and Security of Networks and Services												
01/01/2020 - 31/12/2030												
Pillars												
Smart Government												
Phase One												
01/01/2020 - 31/12/2025												
Smart Cities												
01/01/2021 - 31/12/2025												
Smart Education												
01/01/2021 - 31/12/2025												
Smart Agriculture												
01/01/2021 - 31/12/2025												
Phase Two												
01/01/2022 - 31/12/2030												
Smart Energy												
01/01/2022 - 31/12/2030												
Smart Health												
01/01/2022 - 31/12/2030												
Smart Transport												
01/01/2023 - 31/12/2030												
Smart Trade and Commerce												
01/01/2023 - 31/12/2030												
Smart Tourism												
01/01/2023 - 31/12/2030												
Smart Mining												
01/01/2023 - 31/12/2030												

NB: Some smart solutions will be initiated in between or parallel to the above applications depending on resource availability.

7.3. Project implementation strategy: summary & timeline

The implementation of the smart solutions given below are anchored on the following strong foundation or common *Platforms*:

- Policies, Regulations and Standards;
- Secure and Shared Infrastructure;
- Partnerships, Skills, Capacity Building and Content Development;
- Confidence and Security of networks and services.

SMART SOLUTIONS

PREREQUISITES: Comprehensive SMART Digital Infrastructure countrywide, Security on networks, ICT skills/knowledge to operate and optimally use the technology, Content development, and Appropriate policies and regulatory frameworks.

1	Smart Government (<i>Phase One</i>)				
	Strategic Interventions	Project Activities	Budget (ZWL)	Key Players	Timeline
1.1	E-Cabinet (Paperless Cabinet)	• Requirements gathering (status, user needs, business needs, infrastructure needs) • Design of solution • Procurement of Servers, software, client access devices, and LAN • Implementation of LAN • Develop Software applications (Document management, e-collaboration, email, firewalls, e-voting and e-participation) • Deployment, training and support	\$xxx	• Office of The President and Cabinet • Ministry of ICT, Postal and Courier Services • ICT Sector players	01/12/2021-31/03/2025
1.2	Government Intranet – 5 Ministries for phase 1 • <i>Ministry Foreign Affairs and International Trade including missions,</i> • <i>Ministry of Women Affairs Gender and</i>	• Requirements gathering • Design of solution • Procurement of Servers, software, client access devices, and LAN • Implementation of LAN & WAN • Development of Software (document exchange, Automation of core processes)	\$xxx	• Ministry of ICT, Postal and Courier Services • Office of The President and Cabinet • Government Ministries, Parastatals and State Owned enterprises • ICT Sector Players • Business Communities • Citizens	01/01/2021 - 31/12/2025

	<i>Community Development,</i> • <i>Ministry of Information Communication Technology Postal and Courier Services,</i> • <i>Ministry of Industry and Commerce,</i> • <i>Ministry of Higher and Tertiary Education Science and Technology Development)</i>	• Unified communications (VOIP, chat, email) • Deployment, training and support			
1.3	Government Cloud	• Requirements gathering • Design of solution • Procurement of Servers, software, client access devices, and LAN • Implementation of LAN & WAN • Provision of storage services • Processing Services • Provision of application hosting Services • Provision of Communication Services	\$xxx	• Ministry of ICT, Postal and Courier Services • Office of The President and Cabinet • Government Ministries, Parastatals and State Owned enterprises • ICT Sector Players • Business Communities • Citizens	01/01/2020 - 31/12/2022
1.4	Digitization of Government Content • <i>Ministry of Justice, Legal, and Parliamentary Affairs</i> • <i>Ministry of Industry & Commerce</i>	• Requirements gathering • Design of solution • Procurement of Servers, software, client access devices, and LAN • Implementation of LAN & WAN • Acquisition / Development of Electronic Records & Documents Management System (ERDMS) solutions • Capacity building	\$xxx	• Ministry of ICT, Postal and Courier Services • Office of The President and Cabinet • Government Ministries, Parastatals and State Owned enterprises • ICT Sector Players	01/01/2021 - 31/12/2030

	- Ministry of Finance and Economic Development - Ministry of Home Affairs	- Digitization and conversion of paper records. - Deployment of ERDMS - Transfer of digitized records to National Data Centre (NDC).			
1.5	e-Parliament	- Requirements gathering - Design of solution - Procurement of Servers, software, client access devices, and LAN - Implementation of LAN & WAN - Implement e-Parliament solution - Roll out of e-parliament to constituencies - Link Parliament to the National Data Centre - Provide capacity building for Parliamentarians	\$xxx	- Ministry of ICT, Postal and Courier Services - Ministry of Justice Legal, and Parliamentary Affairs - Office of the President and Cabinet - Government Ministries - Citizens	01/01/2022 - 31/12/2025
2	Smart Agriculture (Phase 1: 300 hectares under irrigation, 150 hectares under natural rainfall, 10-15 hectares for communal farms per province)				
	Strategic Interventions	Project Activities	Budget (ZWL)	Key Players	Timeline
2.1	Smart Infrastructure for Agriculture	- Requirements gathering - Design of solution - Procurement of Servers, software, and devices - Provide Communications Infrastructure	\$xxx	- Ministry of Lands, Agriculture, Water, Climate and Rural Settlement - Ministry of ICT, Postal and Courier Services - Office of the President and Cabinet - Government Ministries	01/01/2021 - 31/12/2025

		• Develop automated solutions for irrigation, water conservation, and farm processes. • Configure smart sensors (Soil scanning, water and moisture, light, humidity, temperature management); • Setup Surveillance systems (IP cameras, alarms, drones)		• Organisations and industries involved in Agriculture • ICT Sector Players • Farmers • Citizens	
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2.2	Smart Information	• Provide Databases for the Agriculture sector – Collection and storage of agriculture data and information, Data inputs, analytics, information dissemination to farmers, AGRITEX officers, decision makers and other stakeholders. • Provide IoT-based solutions		• Ministry of Lands, Agriculture, Water, Climate and Rural Settlement • Ministry of ICT, Postal and Courier Services • Office of the President and Cabinet • Government Ministries • Organisations and industries involved in Agriculture • ICT Sector Players • Farmers • Citizens	
2.3	Smart Logistics	• Implement electronic tracking of transportation of goods • Implement inventory control systems, and • Implement mobile asset tracking		• Ministry of Lands, Agriculture, Water, Climate and Rural Settlement • Ministry of ICT, Postal and Courier Services • Office of the President and Cabinet • Government Ministries • Organisations and industries involved in Agriculture • ICT Sector Players • Farmers • Citizens	

2.4	Software applications	Development of specialised software applications (Smart Agriculture Management System, Automated Decision Systems, Decision support/Knowledge Based Systems, Incident Management, Weather prediction, embedded software)		- Ministry of Lands, Agriculture, Water, Climate and Rural Settlement - Ministry of ICT, Postal and Courier Services - Office of the President and Cabinet - Government Ministries - Organisations and industries involved in Agriculture - ICT Sector Players - Farmers - Citizens	
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3 Smart Education					
	Strategic Interventions	Project Activities	Budget (ZWL)	Key Players	Timeline
3.1	Smart Infrastructure for Education	Develop infrastructure model for smart school (smart energy solution, smart communication, smart building)	\$xxx	- Ministry of Higher and Tertiary Education Science and Technology Development - Ministry of Primary and Secondary Education - Ministry of ICT, Postal and Courier Services - Office of the President and Cabinet	01/01/2021 - 31/12/2025
3.2	ICTs in the classroom	Develop smart classroom model (collaborative tools, digital learning resources, e-library)			

3.3	Smart teaching / smart learning	• Acquire/Develop specialized e-learning software • Provide smart teaching and learning tools (interactive boards, assistive tools, Augmented Reality tools, Virtual Reality tools, etc)		• Other Government Ministries • Education Sector Players • ICT Sector Players • Development Partners • Citizens	
3.4	AI (Artificial intelligence)	• Identify relevant AI solutions • Develop / Acquire AI solutions • Deploy AI solutions			
3.5	Capacity building	• Equip teachers / instructors with requisite skills for smart technology diffusion			

4	Smart City				
	Strategic Interventions	Project Activities	Budget (ZWL)	Key Players	Timeline
4.1	Smart Infrastructure	• Develop smart building model (energy saving tech, IP cameras, smart sensors) • Develop smart urban transport solution • Develop integrated ICT, security and surveillance solution • Identify pilot buildings and streets • Set up pilot Network infrastructure • Equip pilot smart buildings and homes	\$xxx	• Ministry of Local Government, Public Works and National Housing • Ministry of ICT, Postal and Courier Services • Office of the President and Cabinet • Government Ministries • Local Authorities • ICT Sector Players • Citizens	01/01/2022 - 31/12/2030

4.2	Smart Services	• Develop / acquire local government services (e.g. smart health, smart transport, online licenses applications, shop licenses, Vendor licenses, liquor licenses, Residential and commercial land application etc.) • Develop/Deploy smart billing systems. •			
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4.3	Smart People	• Setup public participation platforms (budget and expenditure consultation) • Rollout ICT capacity building			
4.4	Smart Environment	• Setup renewable energy solutions • Develop disaster management systems • Setup smart monitoring solutions (water, soil, and air contamination detection solutions) • Setup waste management solutions			

- NB:
1. Budgetary provisions will be commensurate with the price dictates at the time of implementation.
 2. The above list of smart solutions form the first phase of smart solutions contained in the Smart Zimbabwe Master Plan.
 3. Some Smart Solutions will be adopted and developed in collaboration with stakeholders and development partners depending on available resources from Government, private Sector and other Cooperating Partners.